

3.1 General properties of waves

5054 Paper 2 only, 2015 onwards. Each item is the printed question (with figures) then the official mark-scheme page(s).

2015

May/June 5054/21

Question 4

- 4 Fig. 4.1 shows a wave on a string. The wave is travelling towards the right.

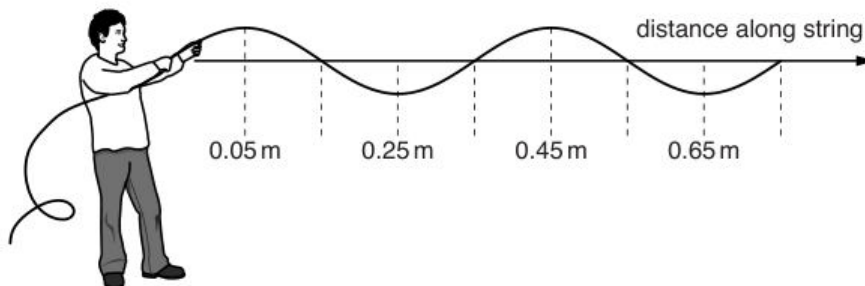


Fig. 4.1

- (a) Describe the movement of the particles of the string.

.....
.....
.....
.....[2]

- (b) Determine the wavelength of the wave.

wavelength =[1]

- (c) The speed of the wave is 2.0 m/s.

- (i) Calculate the frequency of the wave.

frequency =[2]

- (ii) On Fig. 4.1, draw the string at a time 0.10 s later than the time in Fig. 4.1. [1]

Mark scheme — 5054/21 Q4

Page 2	Mark Scheme	Syllabus	Paper
	Cambridge O Level – May/June 2015	5054	21
1	(a) (i) 60 m		B1
	(ii) 12 s		B1
	(b) (i) straight line from origin to 200 m at 40 s any line straight or curved from (40,200) to (60,500)		B1 B1
	(ii) $s = d/t$ or $500/60$ 8.3 m/s		C1 A1
2	(a) (i) force moves through a distance (in same direction)		B1
	(ii) chemical (potential) energy		B1
	(b) (i) 480 Nm		B1
	(ii) attempt to apply moments with two forces and distances 400 N		C1 A1
3	(a) Pa or N/m^2 or cm of mercury or atmosphere(s)		B1
	(b) correct points plotted at $(0.5V_0, 2P_0)$ and $(2V_0, 0.5P_0)$ curve through points of decreasing gradient		B1 B1
	(c) molecules hit sides/piston		B1
	more molecules hit per second/hit more frequently		B1
	molecular impacts create large(r) force (upwards on piston)		B1
4	(a) oscillate/vibrate stated or described transverse movement described		B1 B1
	(b) 0.40 m		B1
	(c) (i) $v = f\lambda$ or $(f =) v/\lambda$ or $2/(b)$ 5.0 Hz		C1 A1
	(ii) clear attempt to draw wave moved along 0.20 m to right		B1
5	(a) $\sin i/\sin r$ or $\sin 50/\sin 30$ 1.5(321)		C1 A1

Page 5	Mark Scheme	Syllabus	Paper
	Cambridge O Level – May/June 2015	5054	21

(iii) 1	$(E =)Pt$ or 200×120 24 000 J	C1 A1
2	$(E =) mcT$ or $100 \times 4.2 \times 20$ 8400 J	C1 A1
3	$(E =) mL$ or 5×2250 11 250 J	C1 A1
4	4350 J or $1 - (2+3)$	B1
11 (a) (i)	51	B1
(ii)	more protons than electrons or different number of protons and electrons positive and negative do not cancel	B1 B1
(iii)	25 protons a different number of neutrons	B1 B1
(b) (i)	147	B1
(ii)	α has mass number 4 α has proton number 2 correct proton number for U ecf their value for α	B1 B1 B1
(c) (i)	alpha particles only travel a short distance in air or alpha particles stopped / scattered / deflected by air or alpha particles ionise air	B1
(ii)	particles come off in different directions or not emitted in one line / as a ray or not all the particles pass through the slit	B1
(iii)	B correct shape and deflected more than A	B1
(iv)	particles close to / fired at the nucleus are deflected (back) / repelled some particles pass (straight) through a few particles come back / large deflection or most pass (straight) through (with little deviation) and how this explains the nucleus is small	B1 B1 B1

© Cambridge International Examinations 2015

Question 9

- 9 Fig. 9.1 shows the wavefronts of a water wave in deep water in a ripple tank.

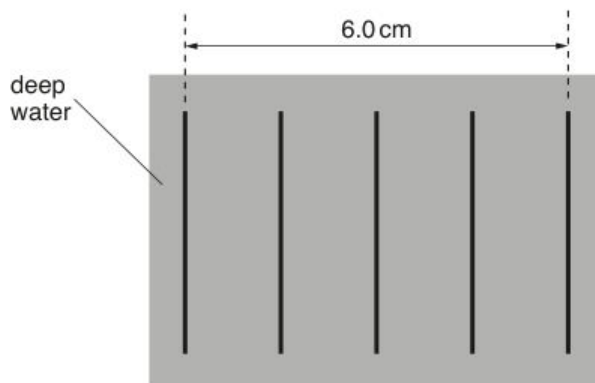


Fig. 9.1

The frequency of the wave in deep water is 5.0 Hz.

- (a) Explain what is meant by the *frequency* of a wave.

.....

 [2]

- (b) (i) Determine the wavelength of the wave in deep water.

..... [1]

- (ii) Calculate the speed of the wave in deep water.

speed = [2]

- (c) The wave passes from deep water into shallow water. The speed of the wave is less in shallower water.

- (i) State and explain how this affects the wavelength of the wave.

.....

 [2]

- (ii) The wave in deep water shown in Fig. 9.2 travels towards the right and enters the shallow water at an angle. The wave refracts.

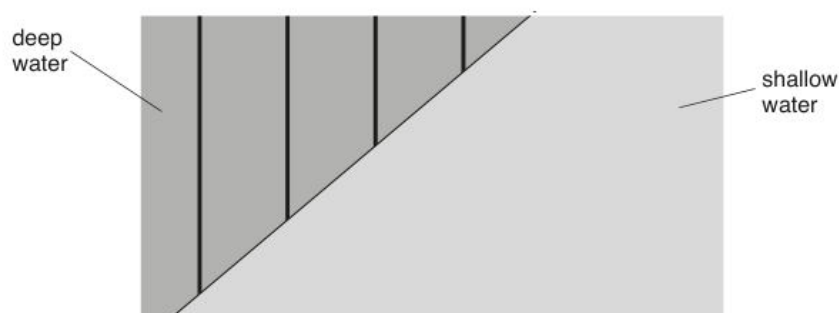


Fig. 9.2

On Fig. 9.2, draw the wavefronts in the shallow water.

[3]

- (d) Sound is also a wave.

- (i) Describe one difference between a sound wave and a water wave.

.....

.....

.....

..... [2]

- (ii) The speed of sound in carbon dioxide gas is less than the speed of sound in air. Using this information, or otherwise, describe an experiment to show the refraction of sound waves. You may include a diagram of your apparatus.

.....

.....

.....

.....

.....

..... [3]

Mark scheme — 5054/22 Q9

Page 4	Mark Scheme	Syllabus	Paper
	Cambridge O Level – May/June 2015	5054	22
7	(a) to provide a complete circuit (with live) or to pass current back to mains or provide a return path for the current (b) current/charge/electrons flow to earth/earth wire/ground (when live touches case) fuse melts/blows and disconnects circuit/cuts live/stops current (c) doubly insulated or case/body made of plastic/insulator/not made of metal or user cannot touch metal (d) (circuit breaker) - turns off/acts fast(er) - can be reset - easy to see it has tripped/switched - can detect small difference between live and neutral currents / small (leakage) current to earth	B1	B1
8	(a) left column both 1 right column 0 and 1 (b) (at least one of the atoms) contain same number of electrons and protons or have 1 electron and 1 proton charge on electron and proton opposite or electron negative and proton positive or charge on electron neutralises/cancels/balances proton charge neutrons have no charge	B1	B1
9	(a) number of waves (that pass a point) or number of oscillations (passing a point) in unit time or per second or in 1 second (b) (i) 1.5 cm (ii) $(v =)f\lambda$ or 5×1.5 seen or $(s=)d/t$ and $f = 1/t$ 7.5 cm/s (c) (i) wavelength decreases travels a shorter distance in the same time or frequency stays the same (and $v = f\lambda$)	M1	A1
		B1	C1
		A1	
		B1	B1

© Cambridge International Examinations 2015

Question 7

7 Audible sound waves and ultrasound waves both consist of compressions and rarefactions.

(a) Describe one difference between a compression and a rarefaction.

.....
.....
..... [1]

(b) In a particular medium, the speed of both audible sound and ultrasound waves is v .

(i) State the equation that relates the wavelength λ of a sound wave to its frequency f .

.....
..... [1]

(ii) Explain how, in this medium, the wavelength of an audible sound wave compares with the wavelength of an ultrasound wave.

.....
.....
.....
..... [1]

(c) Describe one use of ultrasound waves.

.....
.....
.....
.....
..... [2]

Mark scheme — 5054/21 Q7

Page 3	Mark Scheme	Syllabus	Paper
	Cambridge O Level – October/November 2015	5054	21
5	(a) temperature at which a liquid becomes a gas	B1	
	(b) (i) molecules close together / touching or closer than in gas randomly arranged or irregular structure	B1 B1	
	(ii) to separate / increase the distance between molecules work done against (intermolecular) forces or supply p.e. or break bonds	B1 B1	[5]
6	(a) distance from (optical) centre to focal point (principal focus)	B1	
	(b) (i) both Fs correctly positioned at ± 1 mm	B1	
	(ii) two of: paraxial ray to lens through focal point to image ray through optical centre ray through focal point and then paraxial to image (ign. arrows) X at crossing point of rays	M2 A1	
	(iii) 3.4–3.8 cm	B1	[6]
7	(a) at compression: molecules closer together or pressure higher or vice versa for rarefaction	B1	
	(b) (i) $v = f\lambda$ or in words	B1	
	(ii) larger and because the frequency is lower	B1	
	(c) states one use basic idea	(e.g. prenatal scanning) (e.g. ultrasound reflects off foetus)	B1 B1 [5]
8	(a) (i) $(I =)V/R$ or $12/(6000 + 2000)$ or $12/8000$ or $12/2000$ or $12/6000$ or in (ii) $(V =)IR$ or 0.0015×6000 1.5 mA	C1 A1	
	(ii) 9.0 V	B1	
	(b) (reading / it) increases resistance of LDR falls	B1 B1	
	(c) light meter / sensor or automatic light switch or something sensible	B1	[6]
[45]			

Page 4	Mark Scheme	Syllabus	Paper
	Cambridge O Level – October/November 2015	5054	21

Section B

- 9 (a) (i) (vector) has direction **or** scalar does not have direction **or**
(vectors) may cancel **or** scalars cannot cancel B1
- (ii) one vector quantity e.g.: displacement; velocity, acceleration B1
one scalar quantity e.g.: distance, length, speed, time, current, resistance B1 [3]
- (b) downward weight arrow of length 3.9–4.1 cm B1
correct triangle/parallelogram drawn **and** correct diagonal clear B1
270–285 kN B1
horizontal ($\pm 3^\circ$) B1
- (c) (i) from chemical/fuel energy B1
to kinetic (and thermal energy) (**not** with any intermediate) B1
- (ii) **air** resistance/friction/drag B1
air resistance/friction/drag increases **or** resultant force decreases B1
or acceleration decreases B1
resultant force is zero **or** (air) resistance/friction equals thrust B1
direction of motion is changing B1
velocity is vector **or** has a direction B1
(acceleration depends on) changing velocity **or**
resultant force towards centre (of circle) **or** **centripetal** force B1 [12]
- [15]
- 10 (a) (i) at least two straight parallel lines inside the coil B1
at least two (complete) lines one above the coil and one below the coil B1
third line in middle **and** evenly spaced **and** two closed loops B1
(any crossings max. 2/3)
- (ii) current (in X) increases B1
magnetic field becomes stronger/changes B1
current/e.m.f./voltage **induced** in Y/electromagnetic induction B1
opposite deflection B1
larger deflection B1
magnetic field decreasing **or** quicker (rate of) change B1 [9]
- (b) (i) to increase the strength of the magnetic field B1
to direct/concentrate the magnetic field (into the secondary coil) B1
- (ii) $(P =)VI$ **or** $33\,000 \times 85$ C1
 $2.8 \times 10^6 \text{ W}$ **or** 2800 kW **or** 2.8 MW A1
- (iii) $(E =)VIt$ **or** $33\,000 \times 85 \times 3600$ **or** $2.8 \times 10^6 \times 3600$ C1
 $1.0/1.01/1.008 \times 10^{10} \text{ J}$ A1 [6]
- [15]

© Cambridge International Examinations 2015

2016

May/June 5054/21

Question 5

- 5 (a) Light is a transverse wave. The direction of vibration is perpendicular to the direction of transfer of the energy.

Complete the table of Fig. 5.1 to show the direction of vibration and the type of wave associated with a sound wave and with a wave on the surface of water in a ripple tank.

wave	direction of vibration	type of wave
light	perpendicular to the direction of transfer of the energy	transverse
sound		
water		

[2]

Fig. 5.1

- (b) A wave travels along a rope. Fig. 5.2 shows how the height of one point on the rope varies with time as the wave passes.

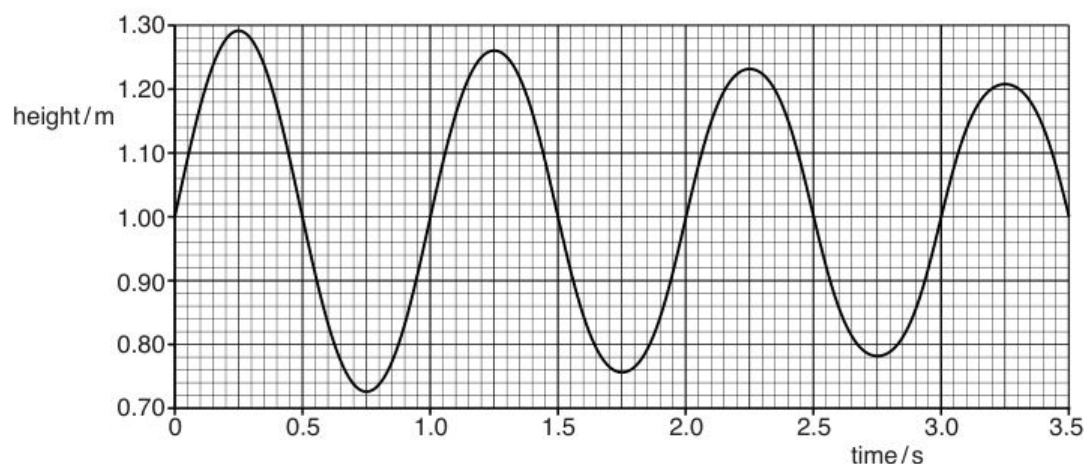


Fig. 5.2

- (i) Determine the amplitude of the vibration at time = 0.25 s.

.....[1]

- (ii) Describe how the graph shows that the frequency of the wave is constant.

.....
[1]

Page 3	Mark Scheme	Syllabus	Paper
	Cambridge O Level – May/June 2016	5054	21
(b) (i) 1	difference between smallest and largest temperature or from 0 to 100 °C		B1
(i) 2	small/moderate distance between (thermometer) marks or for a given temperature change there is a small expansion of liquid / distance (along scale) / change in thermometric property or cannot measure small temperature <u>difference / change</u>		B1
(ii) •	use liquid that expands more • smaller bore / thinner tube • more mercury (in bulb) or use larger bulb		B1
5 (a)	<i>sound</i> : along or parallel (to transfer of energy or wave) and longitudinal <i>water</i> : perpendicular and transverse		B1 B1
(b) (i)	0.29 – 0.28 m		B1
(ii)	<u>time / period</u> for one wave (length) / cycle constant or each oscillation / cycle takes one second		B1
6 (a)	angle of incidence smallest angle for light to be totally internally reflected or largest angle (of incidence) for ray to be refracted / emerge or when light emerges along surface or when angle of refraction is 90°		B1 B1
(b) (i)	$n = 1 / \sin C$ algebraic or numerical 2.5 or 2.46 or 2.458(59)		C1 A1
(ii)	<i>left hand diagram</i> ray refracts away from normal and emerges into air at bottom left surface <i>right hand diagram</i> reflected horizontal ray (by eye) <i>right hand diagram</i> rest of ray completely correct to emerge into air at top face without refraction (by eye)		B1 B1 B1
7 (a)	(current in coil) creates magnetic field or current is at right angles to magnetic field (of permanent / cylindrical magnets)		C1
(b)	into and out of magnet or left and right or backwards and forwards current is one way then reverses (so reverses force)		B1 B1

Page 4	Mark Scheme	Syllabus	Paper
	Cambridge O Level – May/June 2016	5054	21

	(c) ($\lambda =$) v/f numerically or algebraic in any form 0.64 m	C1 A1
8	(a) (i) same/equal or $I_B = I_1 = I_2$ (ii) (p.d. of) battery is sum of (p.d. across) fixed resistor and (p.d. across) the variable resistor or $V_B = V_1 + V_2$	B1 B1
	(b) ($I =$) V/R numerical or algebraic in any form 0.006(0) A	C1 A1
9	E (a) 2 squares 10 V	C1 A1
	(b) measure/find horizontal distance/number of divisions (between points) distance $\times$ no (m)s/division	C1 A1
	OR (a) transistor	B1
	(b) (in dark) resistance of LDR large/increases large voltage across base (and emitter) switches transistor on or current in <u>collector</u> increases	B1 B1 B1
10	(a) (i) temperature when solid turns to liquid (ii) molecules escape (surface) fastest molecules/most energetic molecules escape/break bonds leaving behind slower molecules/colder molecules or temperature falls	B1 B1 C1 A1 B1
	(b) (i) at the surface/top of liquid (ii) less heat/energy <u>enters</u> (liquid nitrogen)/transfers or less nitrogen evaporates/boils reduces/stops conduction and convection explanation of no conduction or convection, e.g. no molecules/no medium	B1 B1 B1 B1

© Cambridge International Examinations 2016

Question 5

- 5 Fig. 5.1 shows a loudspeaker inside a glass bell-jar that contains air.

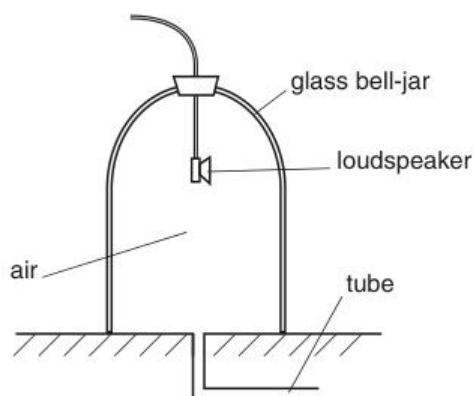


Fig. 5.1

The loudspeaker produces a sound of frequency 2200 Hz.

- (a) State what is meant by the *frequency* of a sound.

.....

[1]

- (b) (i) The speed of sound in air is 330 m/s.

Calculate the wavelength of the sound in the air.

wavelength =[2]

- (ii) The sound passes from the air into the glass.

State what happens to

1. the frequency of the sound,

.....

2. the speed of the sound.

.....

[1]

(c) (i) Describe

1. how sound is produced by the loudspeaker,

.....
.....
.....[1]

2. how sound is transmitted through the air.

.....
.....
.....[2]

- (ii) A pump is connected to the tube at the bottom of the glass bell-jar. The pump is switched on and it removes the air from the glass bell-jar.

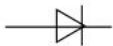
Explain why the volume of the sound heard outside the glass bell-jar decreases.

.....
.....
.....[1]

Page 3	Mark Scheme	Syllabus	Paper
	Cambridge O Level – October/November 2016	5054	21
5	(a) number of oscillations / vibrations / wavelengths / compressions / rarefactions / cycles per second / unit time	B1	
	(b) (i) $(\lambda =)c/f$ or 330 / 2200 0.15 m	C1 A1	
	(ii) 1 no change and 2 increases	B1	
	(c) (i) 1 loudspeaker vibrates / oscillates / moves to and fro (and collides with molecules) 2 compressions and rarefactions / molecules vibrate / longitudinal wave vibration / oscillation / energy passed on	B1 B1 B1	
	(ii) fewer / no molecules / particles and less / no energy / vibration transferred	B1	[8]
6	(a) (i) X N-pole Y S-pole and Z N-pole	B1 B1	
	(ii) they touch / move towards each other and opposite poles attract	B1	
	(b) any sensible use: starting-motor circuit; with a logic gate; nuclear power station corresponding explanation: current too large for dash-board switch; current too small to power device; too dangerous to reach switch	B1 B1	[5]
7	(a) (i) supplies the (mains) e.m.f. / voltage	B1	
	(ii) to complete the circuit / is at 0 V	B1	
	(b) (i) the circuit / supply is cut / broken or current stops fuse melts / blows / burns	B1 B1	
	(ii) live wire when it cuts the circuit / melts no part of the appliance is live / no shock	B1 B1	[6]
[45]			

Page 4	Mark Scheme	Syllabus	Paper
	Cambridge O Level – October/November 2016	5054	21

Section B

- 8 (a) (i) 11 protons **and** 11 electrons B1
13 neutrons B1
electrons in orbit/surrounding nucleus **or** neutrons **and** protons in nucleus B1
- (ii) **one** more neutron (in sodium-24) **or** one fewer neutron in sodium-23 B1 [4]
- (b) (i) electron B1
- (ii) ${}_{-1}^0(\beta)$ **cao** B1
 ${}_{12}^{24}(\text{Mg})$ B1
 ${}_{12}^{24}(\text{Mg})$ B1 [4]
- (c) electromagnetic (radiation / rays / waves) M1
(very) high frequency / energy **or** (very) short wavelength A1 [2]
- (d) (i) path curving upwards B1
- (ii) path continues in straight line B1
- (iii) beta-particle charged **or** gamma-ray uncharged B1 [3]
- (e) long enough **or** short enough B1
to take measurements **or** so the body is not irradiated for long B1 [2]
[15]
- 9 (a) (i) magnetic field mentioned B1
alternating / changing magnetic field B1
current / voltage / e.m.f. induced (in secondary coil) B1
- (ii)  B1
diode B1 [5]
- (b) (i) work done / energy transferred per unit charge M1
electrical energy to other forms **or** for whole circuit **or** property of supply A1

Question 10

14

- 10 (a) Fig. 10.1 shows two rays from an object that is placed in front of a plane mirror.

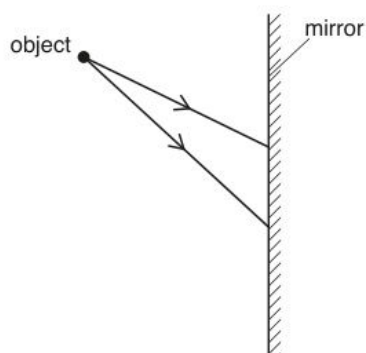


Fig. 10.1

- (i) On Fig. 10.1, draw the two reflected rays and locate the position of the image. [2]
- (ii) Apart from its position, state one characteristic of the image.
[1]
- (b) Fig. 10.2 shows wavefronts in a ripple tank. They move in the direction of the arrow.

The wave hits the boundary between two regions and the wave slows down as it enters the shaded region.

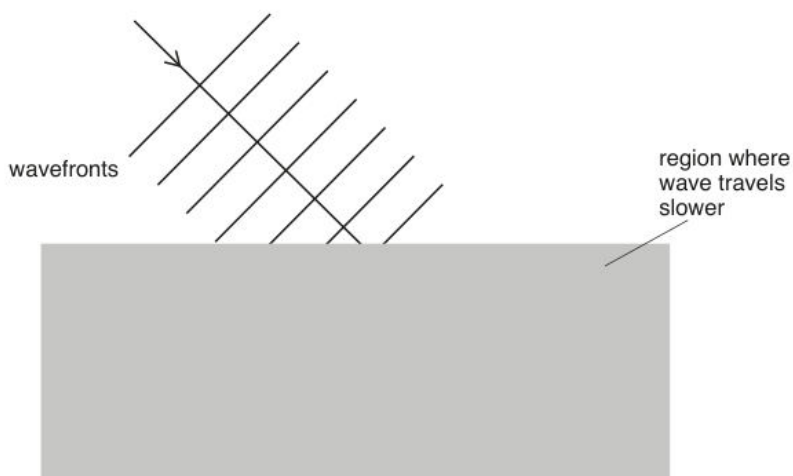


Fig. 10.2

- (i) State what is meant by *wavefront*.

[2]

(ii) Both reflection and refraction occur at the boundary.

1. On Fig. 10.2, draw the reflected wavefronts. [1]

2. On Fig. 10.2, draw the refracted wavefronts. [2]

(c) A loudspeaker produces a sound of frequency 2.0 kHz. The wavelength of this sound in air is 16 cm.

(i) Calculate the speed of sound in air.

speed = [2]

(ii) 1. State the range of frequencies that can be heard by a healthy human ear.

..... [1]

2. Calculate the smallest wavelength of sound that can be heard by a healthy human ear.

wavelength = [1]

(iii) Describe a simple experiment to show that sound waves obey the law of reflection.
You may draw a diagram if you wish.

.....
.....
.....
..... [3]

Mark scheme — 5054/21 Q10

5054/21

Cambridge O Level – Mark Scheme
PUBLISHED

May/June 2017

Question	Answer	Marks
10(a)(i)	both reflected rays correct by eye	B1
	image in correct position shown by continuation of rays behind mirror	B1
10(a)(ii)	virtual or upright or laterally inverted or same size (as object)	B1
10(b)(i)	line (joining points)	C1
	line joining points / particles on wave with same phase or line joining points along a crest etc.	A1
10(b)(ii)1	correct angle to surface $\pm 7^\circ$ with correct orientation and similar wavelength (by eye)	B1
10(b)(ii)2	at least two lines at smaller angle to surface with correct orientation in slower medium	B1
	showing smaller and constant wavelength with wavefronts deviated in correct direction	B1
10(c)(i)	$(v =) f\lambda$ or $2(000) \times 16$	C1
	32 000 cm / s or 320 m / s	A1
10(c)(ii)1	a range with 15–25 Hz as the lowest frequency and 15–30 kHz as the highest	B1
10(c)(ii)2	1.6 cm (if highest frequency is 20 000 Hz)	B1
10(c)(iii)	tube or other method to produce (narrow) beam of sound on source and/or on detector	B1
	stated detector or stated reflector	B1
	detector moved to find maximum loudness or angles i and r measured with suitable experiment	B1

© UCLES 2017

Page 7 of 8

May/June 5054/22

Question 6

- 6 (a) Explain the difference between a longitudinal wave and a transverse wave.

You may draw a diagram, if you wish, to help your explanation.

.....

.....

.....

.....

.....[2]

- (b) A sound wave has a frequency of 3.8 kHz and a speed of 330 m/s.

- (i) Calculate the wavelength of the sound.

wavelength =[2]

- (ii) A different sound has a frequency of 3.8 Hz.

State and explain whether a human with normal hearing is able to hear this sound.

.....

.....

.....[1]

Mark scheme — 5054/22 Q6

5054/22

Cambridge O Level – Mark Scheme
PUBLISHED

May/June 2017

Question	Answer	Marks
6(a)	longitudinal - vibration / oscillation / movement to and fro and in direction of wave or has compressions and rarefactions	B1
	transverse – vibration / oscillation / movement up and down and at right angles to wave or has crests and troughs	B1
6(b)(i)	$(\lambda =) v / f$ or 330 / 3800	B1
	0.087 m or 8.7 cm	B1
6(b)(ii)	not heard and because below the range of audible frequencies or audible range is 20 – 20 000 Hz or too low a pitch / frequency	B1

Question	Answer	Marks
7(a)(i)	current in coil (at right angles) in a magnetic field (of magnet) or left-hand rule mentioned	B1
7(a)(ii)	reverses / changes direction of current (in coil)	B1
	reverses current every half turn / when coil is vertical or reverses forces (on side AB / CD) or keeps forces in same direction for wire on one side	B1
7(b)(i)	$(E =) VIt$ or $2 \times 12 \times 8$ or $E = Pt$ and $P = VI$ or $E = VQ$ and $Q = It$	C1
	190 or 192 J	A1
7(b)(ii)	73% or 0.73	B1

© UCLES 2017

Page 5 of 10

Oct/Nov 5054/22

Question 6

- 6 Fig. 6.1 shows a girl standing a few hundred metres in front of a large building.



Fig. 6.1 (not to scale)

The girl uses a signal generator and a loudspeaker to send a short pulse of sound towards the building. The sound has a frequency of 3700 Hz.

A short time later, the girl hears an echo.

- (a) State what is meant by the term *echo*.

.....
.....[1]

- (b) The pitch of the echo is the same as that of the original sound but the echo is not as loud.

State what has happened to

- (i) the amplitude of the sound wave,

.....[1]

- (ii) the frequency of the sound wave.

.....[1]

- (c) The speed of sound in air is 330 m/s.

Calculate the wavelength of this sound.

wavelength =[2]

Mark scheme — 5054/22 Q6

5054/22

Cambridge O Level – Mark Scheme
PUBLISHED

October/November
2017

Question	Answer	Marks
5(a)	joining together of (small) <u>nuclei</u> (to make bigger nuclei)	B1
	energy released	B1
	hydrogen (used) or helium (produced)	B1
5(b)	electromagnetic <u>radiation</u> / infra-red / light / ultraviolet	B1
	travels through vacuum or no medium needed	B1

Question	Answer	Marks
6(a)	reflection of <u>sound</u>	B1
6(b)(i)	decreases	B1
6(b)(ii)	does not change	B1
6(c)	$(\lambda =) c/f$ or 330 / 3700	C1
	0.089 m or 8.9 cm or 89 mm	A1

© UCLES 2017

Page 5 of 10

2018

May/June 5054/21

Question 11

11 A water wave is produced in a ripple tank.

- (a) (i) Draw a diagram of a ripple tank and describe how it is used to produce and observe the wave.

.....

.....

.....

.....

.....

.....

.....[4]

- (ii) The water wave in the ripple tank transfers energy.

Describe how to show that there is a transfer of energy.

.....

.....

.....

.....[2]

- (iii) State what is meant by a *wavefront*.

.....

.....

.....[2]

- (b) Fig. 11.1 shows the variation with time t of the height of water at the same place in the tank for two waves P and Q. The height is sometimes above and sometimes below the mean level which is marked as 0.

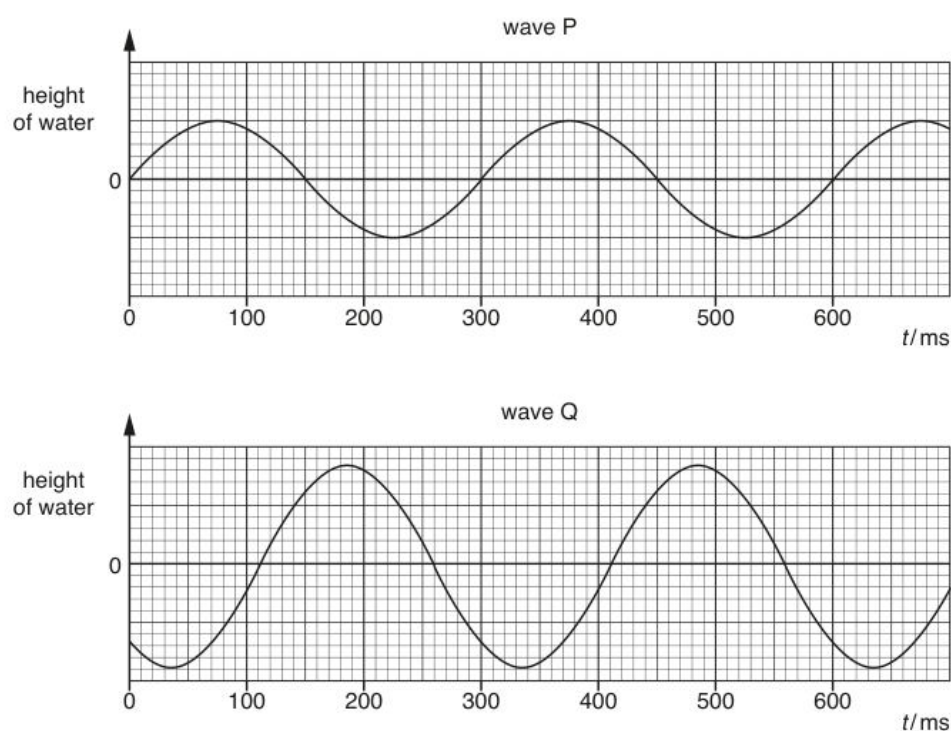


Fig. 11.1

- (i) State one difference and one similarity of the two waves.

difference

.....

similarity

.....

[2]

- (ii) The speed of the water wave is 0.20 m/s.

Determine

1. the frequency of wave P,

frequency =[3]

2. the wavelength of wave P.

wavelength =[2]

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge International Examinations Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cie.org.uk after the live examination series.

Cambridge International Examinations is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.

Mark scheme — 5054/21 Q11

5054/21

Cambridge O Level – Mark Scheme
PUBLISHED

May/June 2018

Question	Answer	Marks
11(a)(i)	tank with water	B1
	dipper (bar or point) touches or in surface of water	B1
	light source / stroboscope / video	B1
	dipper moves up and down (to create wave)	B1
11(a)(ii)	observe (surface of) water OR place particle / cork on surface OR equivalent movable / rotating object	B1
	moves up and down	B1
11(a)(iii)	line joining points	M1
	that are at the same point in the motion	A1
11(b)(i)	<i>difference</i> : Q has a greater / different amplitude or phase	B1
	<i>similarity</i> : same frequency; same time for one wave (period); both transverse; oscillation up and down	B1
11(b)(ii)	time 300 (ms) or 0.3 (s)	C1
	(f=) $1/t$ numerical or algebraic in any form	C1
	3.3 Hz	A1
11(b)(ii)	$2 (\lambda =) v/f$ or $v = \lambda f$ – numerical or algebraic in any form	C1
	0.060 m	A1

May/June 5054/22

Question 5

2

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

- 1 A student drops a small metal object into a cylinder of oil. The object falls alongside a vertical ruler and a camera records its position at 1.0 s intervals, as shown in Fig. 1.1.

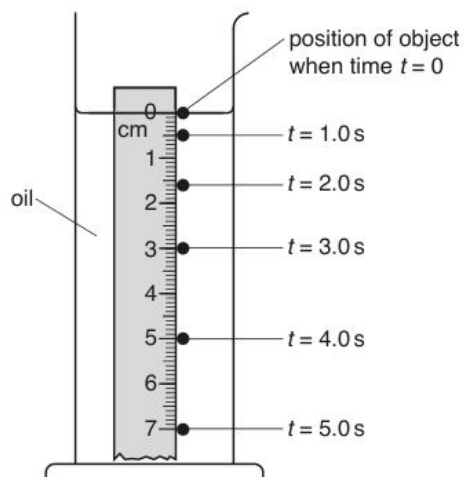


Fig. 1.1

- (a) On the grid in Fig. 1.2, plot a distance-time graph for the object.

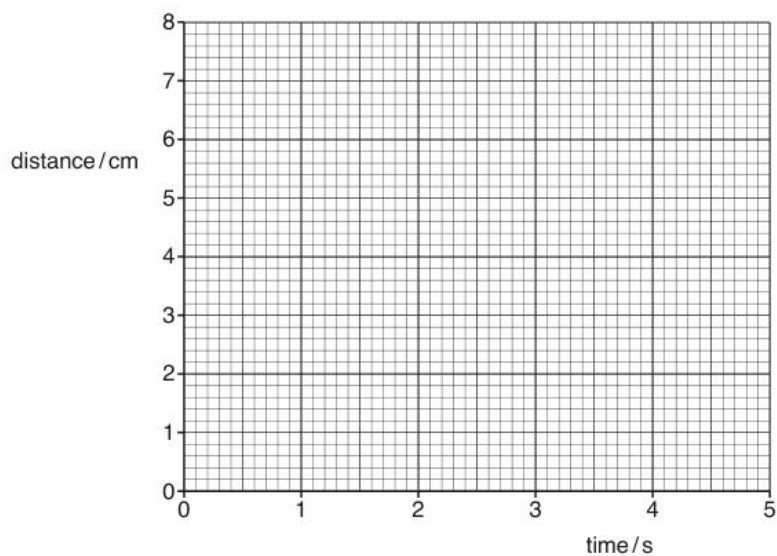


Fig. 1.2

[2]

(b) The object has a constant speed for some of the time.

(i) State and explain how a distance-time graph shows that speed is constant.

.....
.....
.....
.....[2]

(ii) Explain, in terms of the forces acting on the object, how it is able to fall at a constant speed.

.....
.....
.....
.....[2]

- 2 In an experiment, a student takes measurements and determines the extension of a spring for different loads. The apparatus is shown in Fig. 2.1.

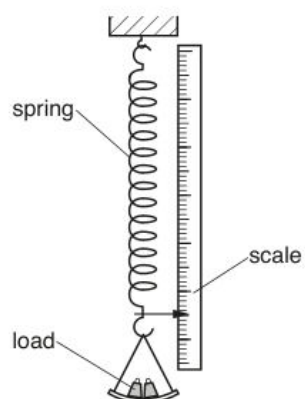


Fig. 2.1

The table in Fig. 2.2 shows the results.

load / N	0	1.0	2.0	3.0	4.0	5.0
length of spring / mm	200	235	270	305	350	420
extension / mm	0	35	70	105		

Fig. 2.2

- (a) Only some of the extensions are shown in the table.

Complete the table to show all of the extensions.

[1]

- (b) Calculate the load that produces an extension of 49 mm.

load =[2]

- (c) The student pulls a load downwards from position A to position B and holds it fixed at position B, as shown in Fig. 2.3.

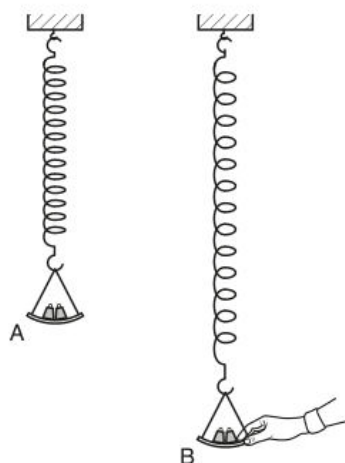


Fig. 2.3

The load is stationary at A and at B. The load has no kinetic energy at either point.

- (i) Place ticks (✓) in the boxes to show how the value of each of the forms of energy compares at A and B.

	equal at A and B	larger at A	larger at B
energy stored in spring	☐	☐	☐
gravitational potential energy of load	☐	☐	☐

[1]

- (ii) Work is done by the student to pull the load down. The law of conservation of energy states that energy cannot be created and cannot be destroyed.

Explain how this principle applies in this case.

.....

.....

.....

.....[2]

3 Equal volumes of metal, water and air are heated from 20 °C to 80 °C.

(a) State whether the metal, the water or the air expands the most.

.....[1]

(b) Fig. 3.1 is a diagram of the metal at 20 °C. The metal is a large, square sheet with a square hole at its centre. The sheet lies flat on a table.

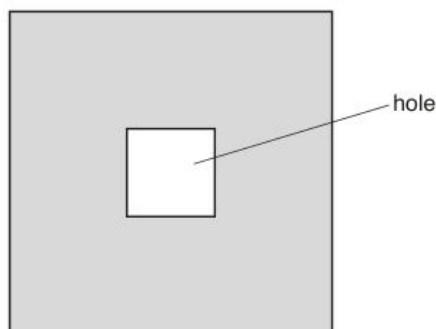


Fig. 3.1

On Fig. 3.1, draw the outline of the sheet and of the hole after the metal is heated. Make sure that the expansion is noticeable. [2]

(c) The metal has a mass of 5.0 kg and a density of $7.5 \times 10^3 \text{ kg/m}^3$.

(i) Define *density*.

.....
[1]

(ii) Calculate the volume of the metal.

Give your answer to a suitable number of significant figures.

volume = [2]

Mark scheme — 5054/22 Q5

5054/22

Cambridge O Level – Mark Scheme
PUBLISHED

May/June 2018

Question	Answer	Marks
5(a)	(d=) s × t in any form algebraic or numerical,	C1
	3.6×10^7 m	A1
5(b)(i)	1st two columns correct (sound longitudinal and microwaves transverse)	B1
	3rd column correct microwaves electromagnetic	B1
5(b)(ii)	Layers / molecules / particles / spring coils close together or high(er) pressure (than atmospheric)	B1

Question	Answer	Marks
6(a)	any two of - ray <u>through</u> middle of lens undeviated - ray parallel to axis passes <u>through</u> focus, 3 cm from lens - ray <u>through</u> focus on left of lens parallel to axis after lens	B2
6(b)	inverted or real	B1
6(c)	image size / object size or image distance / object distance numerical or algebraic	C1
	rays from lens intersect on diagram (for image) and 1.3–1.7	A1
6(d)	projector or photographic enlarger	B1

© UCLES 2018

Page 6 of 10

Oct/Nov 5054/21

Question 9

- 9 Some waves are longitudinal and some waves are transverse.

(a) State how a longitudinal wave differs from a transverse wave.

.....
 [1]

(b) A vibrating rod produces a water wave in a ripple tank. Fig. 9.1 shows the crests of the wave passing into the right-hand section of the tank where the depth of the water is different from the depth in the rest of the tank.

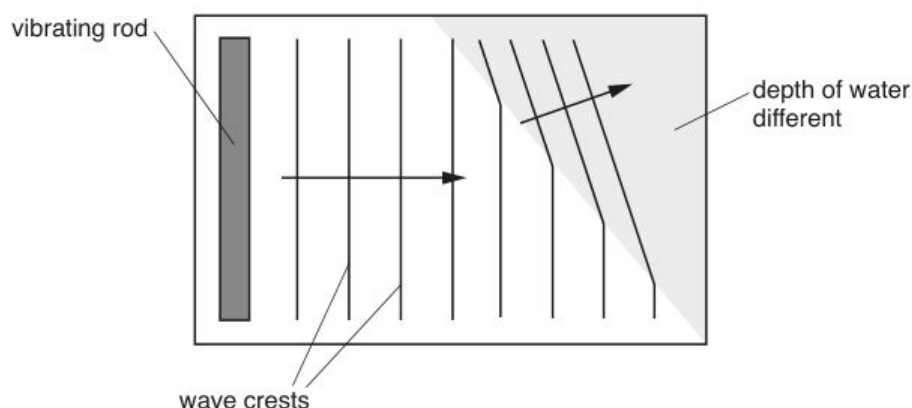


Fig. 9.1

The arrows on Fig. 9.1 show the direction of travel of the wave in the two sections of the ripple tank.

- (i) In the left-hand section of the tank, the wavelength of the wave is 0.019 m and it travels at 0.17 m/s.

Calculate the frequency of the wave.

frequency = [2]

- (ii) State what happens to the frequency of the wave, as it passes into the right-hand section of the tank.

..... [1]

- (iii) Using Fig. 9.1, state and explain what happens to the speed of the wave as it passes into the right-hand section of the tank.

.....

 [2]

- (c) Fig. 9.2 shows light, in air, striking the vertical side of a rectangular glass block at an angle of incidence of 60° .

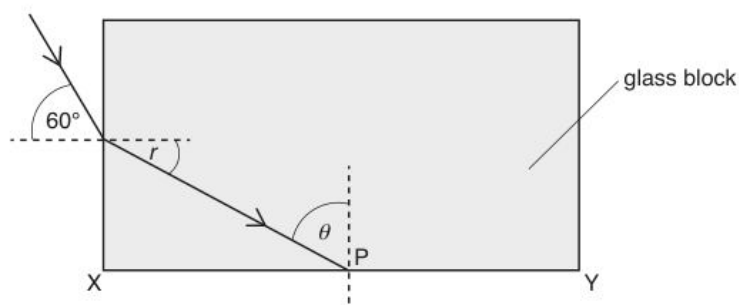


Fig. 9.2

The refractive index of the glass is 1.6. The light travels in the glass and strikes side XY at P.

- (i) Underline all the terms that describe a light wave.

electromagnetic

longitudinal

transverse

[1]

- (ii) At the point where the light enters the glass, the angle of refraction is r .

Calculate angle r .

$r = \dots\dots\dots$ [2]

- (iii) 1. Calculate the critical angle c for light travelling in the block.

$c = \dots\dots\dots$ [2]

2. At P, the angle θ between the ray and the normal is given by $\theta = 90^\circ - r$.

State and explain what happens to the light when it strikes side XY.

.....

 [2]

3. On Fig. 9.2, draw the path of the light after it strikes side XY at P and the path of the light when it is again travelling in the air. [2]

Mark scheme — 5054/21 Q9

5054/21

Cambridge O Level – Mark Scheme
PUBLISHED

October/November 2018

Question	Answer	Marks
8(d)(ii)	from chemical (energy) or chemical (energy) as first term	B1
	to gravitational potential (energy) or to thermal / heat (energy) as last term	B1
	to thermal / internal (energy) / heat and thermal / internal (energy) / heat	B1
8(e)	no resultant force or forces balance / cancel or tension equals weight (of bucket) or upwards force equals downward force	B1

Question	Answer	Marks
9(a)	the <u>vibration</u> direction is parallel to the wave / energy travel direction or longitudinal waves have compressions and rarefactions or transverse waves have crests and troughs or longitudinal waves cannot be polarised	B1
9(b)(i)	$(f =) v / \lambda$ or 0.17 / 0.019	C1
	8.9 Hz	A1
9(b)(ii)	(frequency) stays constant or does not change	B1
9(b)(iii)	(speed) decreases	B1
	(Fig. 9.1 shows that) wavelength decreases or refraction is towards normal or $i > r$ or top of wave (in shaded region) lags behind bottom of wave	B1
9(c)(i)	electromagnetic and transverse underlined	B1
9(c)(ii)	$n = \sin(i) / \sin(r)$ or $(r =) \sin^{-1}(\sin(i) / n)$ or $(r =) \sin^{-1}(\sin(60^\circ) / 1.6)$	C1
	33(°)	A1

© UCLES 2018

Page 8 of 10

Oct/Nov 5054/22

Question 9

9 Both sound and ultrasound are waves that travel in solids, liquids and gases.

(a) State how sound differs from ultrasound.

.....
..... [1]

(b) Sound and ultrasound are longitudinal waves that consist of compressions and rarefactions.

(i) Explain what is meant by a wave and describe how a longitudinal wave differs from a transverse wave.

.....
.....
.....
..... [3]

(ii) Fig. 9.1 represents a longitudinal wave of frequency 25 000 Hz travelling in oil.



Fig. 9.1 (full scale)

1. On Fig. 9.1, mark two points at the centre of two different rarefactions and label each one R. [1]
2. On Fig. 9.1, draw a double-headed arrow to indicate a distance that is equal to one wavelength of the wave. [1]
3. Measure the length of the arrow drawn on Fig. 9.1 and use it to determine the speed of the wave in oil.

speed = [2]

(c) A sound wave travelling in a liquid, passes into air.

(i) State what happens to the speed of the sound wave as it enters the air.

..... [1]

- (ii) The wave in the liquid travels towards the surface at an angle. Fig. 9.2 shows the centres of the compressions of the sound wave in the liquid.

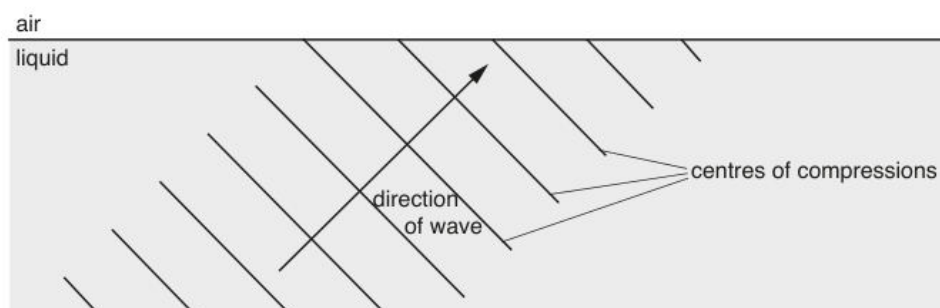


Fig. 9.2

Some compressions shown have reached the liquid-air boundary. The parts of these compressions in the air are not shown on Fig. 9.2.

On Fig. 9.2, draw the parts of these compressions that are in the air. [3]

- (d) Describe, in outline, how ultrasound is used in cleaning.

.....

.....

.....

.....

..... [3]

Mark scheme — 5054/22 Q9

5054/22

Cambridge O Level – Mark Scheme
PUBLISHED

October/November 2018

Question	Answer	Marks
8(c)	any two from: smaller time to drop to zero velocity / hit ground line not straight or velocity does not change uniformly or gradient not constant smaller area under (first part of) graph or less distance travelled slower final velocity initial downward gradient steeper	B2

Question	Answer	Marks
9(a)	frequency of sound wave small(er) or its frequency is less than 20 000 Hz	B1
9(b)(i)	transmission of energy	B1
	(through a medium) with no net movement of medium or by vibrating particles	B1
	vibrations parallel (and antiparallel) to wave / energy travel direction or cannot be polarised	B1
9(b)(ii)	1 two centres of rarefactions labelled <i>R</i>	B1
	2 distance from one point to adjacent identical point indicated (with double-headed arrow)	B1
	3. ($v =$) $f\lambda$ or $25\,000 \times 0.047 / 0.048 / 0.049$ or $25\,000 \times 4.7 / 4.8 / 4.9$ or $25\,000 \times 47 / 48 / 49$	C1
	1200 m / s	A1
9(c)(i)	decreases	B1
9(c)(ii)	four / five straight lines in air that touch the compressions still in the liquid and no intermediate / extra lines between the correct lines	B1
	at least four compressions in air parallel to each other	B1
	at least four straight lines at shallower angle from horizontal and slope correct	B1

© UCLES 2018

Page 8 of 10

2019

May/June 5054/21

Question 10

10 In Fig. 10.1 the vertical lines represent the crests of a wave in a ripple tank.

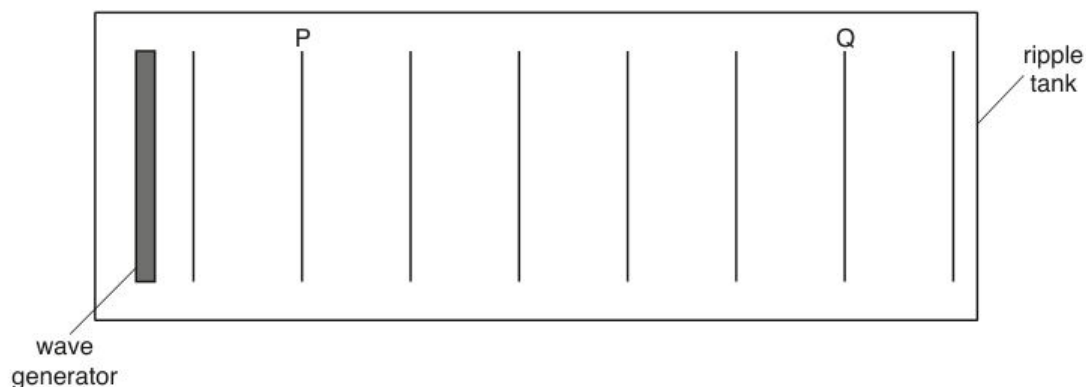


Fig. 10.1

Fig. 10.1 is drawn to scale, with 1.0 cm on the diagram representing 4.0 cm in the tank.

A crest takes 2.0 s to travel from P to Q.

(a) (i) Determine the wavelength of the wave in the ripple tank.

wavelength = [2]

(ii) Define the term *frequency*.

.....
..... [2]

(iii) Determine the frequency of the wave.

frequency = [1]

- (iv) The wave generator is a bar of wood.

Describe how it is used to produce the wave shown.

.....

.....

..... [2]

- (v) Some water in the ripple tank is removed. The water becomes less deep. No other change is made.

State how the change in depth affects the wavelength and the speed of the wave.

speed

wavelength [1]

- (b) A student reads in a textbook that the speed of sound in air is about 300 m/s.

He plans an experiment to check this value.

He asks another student to stand a long distance away in a large open space.

One student has a starting pistol. The pistol, when fired, produces a loud sound and a puff of smoke at the same instant.

- (i) Other pieces of apparatus are needed for this experiment in order to measure the speed of sound.

Name two other pieces of apparatus that are needed.

1.

2. [2]

- (ii) Describe how all the measurements are made.

.....

.....

.....

.....

.....

.....

..... [3]

(iii) State typical values for the speed of sound in:

1. a liquid

.....

2. a solid.

.....

[2]

[Total: 15]

Mark scheme — 5054/21 Q10

5054/21

Cambridge O Level – Mark Scheme
PUBLISHED

May/June 2019

Question	Answer	Marks
10(a)(i)	1.7 (cm) seen or any multiple of wavelength measured on diagram	C1
	6.4–7.2 cm	A1
10(a)(ii)	number of waves/cycles/oscillations in one second / unit time	C1
	number of wavelengths/crests/troughs/ <u>wave</u> cycles/ <u>wave</u> oscillations generated/made/pass a point in one second/unit time	A1
	or number of oscillations of a <u>source/particle</u> in one second / unit time	
10(a)(iii)	2.5 Hz	B1
10(a)(iv)	moved up and down	B1
	regularly or at constant frequency / 2.5 times a second	B1
10(a)(v)	<i>speed</i> smaller / decreases and <i>wavelength</i> smaller / decreases	B1
10(b)(i)	tape measure or trundle wheel or microphones	B1
	stopwatch or timer	B1
10(b)(ii)	start timer on seeing the smoke from gun / sound picked up by microphone	B1
	stop timer on hearing sound	B1
	measure (large) distance between students	B1
10(b)(iii)	both between and including 1000 and 10 000 m/s	B1
	solid speed > liquid speed	B1

Question 5

- 5 Both sound and ultrasound are longitudinal waves.

(a) Describe what is meant by *longitudinal*.

.....

.....

..... [2]

(b) Fig. 5.1 shows an ultrasound cleaner being used to clean jewellery.

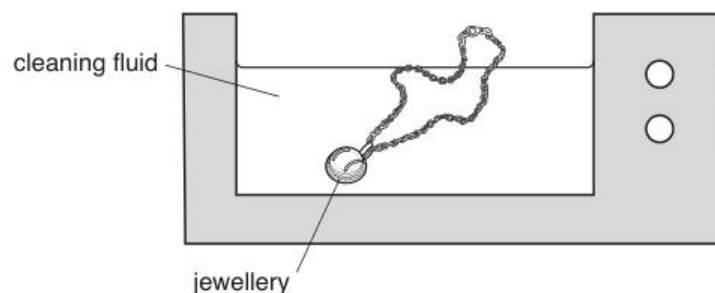


Fig. 5.1

The jewellery is lowered into the cleaning fluid and ultrasound waves of frequency 42 000 Hz are produced in the fluid.

(i) The speed of ultrasound in the fluid is 1500 m/s.

Calculate the wavelength of the ultrasound in the fluid.

wavelength = [2]

(ii) Describe how the ultrasound cleans the jewellery.

.....

.....

..... [2]

[Total: 6]

Mark scheme — 5054/22 Q5

5054/22

Cambridge O Level – Mark Scheme
PUBLISHED

October/November 2019

Question	Answer	Marks
3(a)(i)	(nuclear) <u>fusion</u> (reactions) or nuclei fuse	B1
	small <u>nuclei</u> / hydrogen <u>nuclei</u> joining together (and release energy) or larger <u>nucleus</u> produced	B1
3(a)(ii)	infra-red (radiation) or ultraviolet (radiation) or (visible) light	B1
3(b)	black surfaces are (good) absorbers	M1
	of (infra-red) radiation / heat / (visible) light / thermal energy	A1

Question	Answer	Marks
4(a)	any two from: only occurs at the surface / no bubbles produced occurs at any temperature produces cooling	B2
4(b)	molecules escape (from liquid)	B1
	fast(est) molecules / molecules with great(est kinetic) energy escape or slow(est) molecules / molecules with small(est kinetic) energy remain or (average) speed of remaining molecules decreases or molecules gain (kinetic) energy and escape	B1
4(c)(i)	rate of evaporation increases (with increasing area)	B1
4(c)(ii)	rate of evaporation decreases with decreasing temperature or takes too much time	B1

Question	Answer	Marks
5(a)	vibration / oscillation of <u>particles</u> or energy transfer or compressions and rarefactions	B1
	vibration / oscillation (of particles) parallel to energy travel direction / propagation direction	B1

© UCLES 2019

Page 5 of 10

2021

May/June 5054/22

Question 10

16

- 10 (a) Fig. 10.1 shows a water wave moving from shallow into deep water. The wavefronts shown represent the crests of the wave.

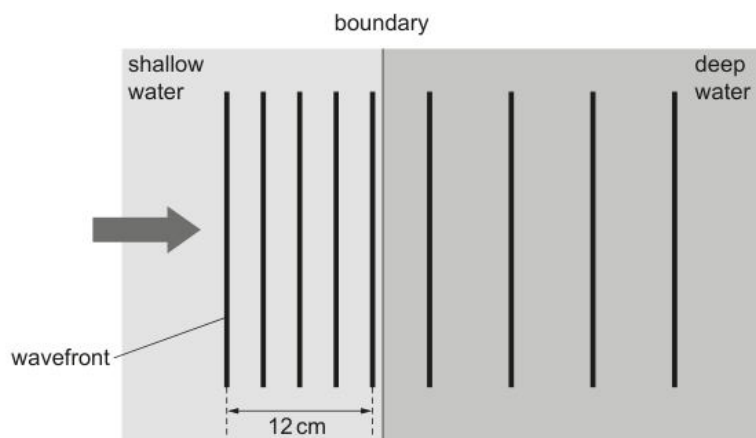


Fig. 10.1 (not to scale)

The water wave is made by dipping a wooden bar up and down in the water. The bar makes 10 complete up and down movements in 5.0 s.

- (i) Determine the wavelength of the wave in the shallow water.

wavelength = [1]

- (ii) Calculate the frequency of the wave produced.

frequency = [2]

- (iii) Determine the speed of the water wave in the shallow water.

speed = [2]

- (iv) On Fig. 10.2, draw lines to show what happens to the wavelength, the frequency and the speed of the water wave as it enters the deep water.

One line has been drawn for you.

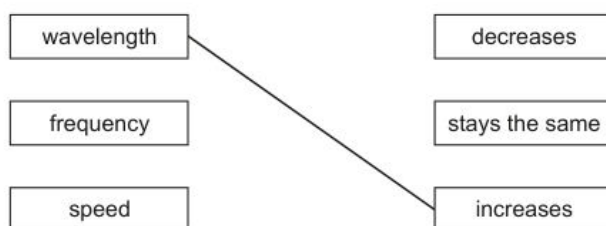


Fig. 10.2

[1]

- (v) The direction of the boundary is now altered. The water wave now enters the deep water from the shallow water at an angle, as shown in Fig. 10.3.

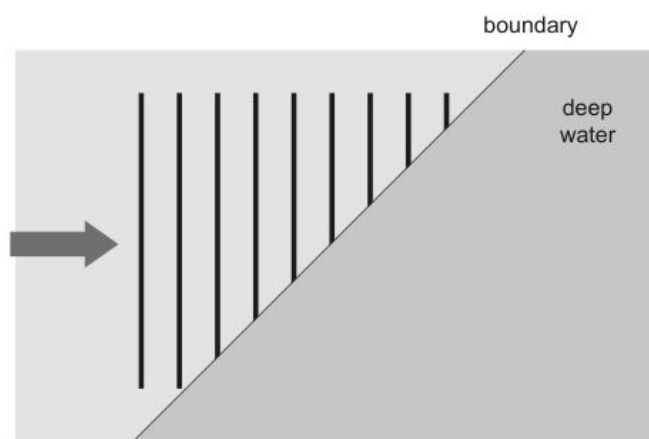


Fig. 10.3

On Fig. 10.3, draw the wavefronts in the deep water.

[3]

(b) White light is made up of different colours.

- (i) State the name of four of the colours in the visible spectrum and place them in order from the smallest wavelength to the largest wavelength.

smallest wavelength

.....

.....

largest wavelength

[2]

- (ii) A narrow beam of white light can be split into different colours.

Fig. 10.4 shows rays of white light emitted from a lamp.

Complete Fig. 10.4 to show how a narrow beam is produced from these rays and how a spectrum is shown on the screen. Label your diagram.

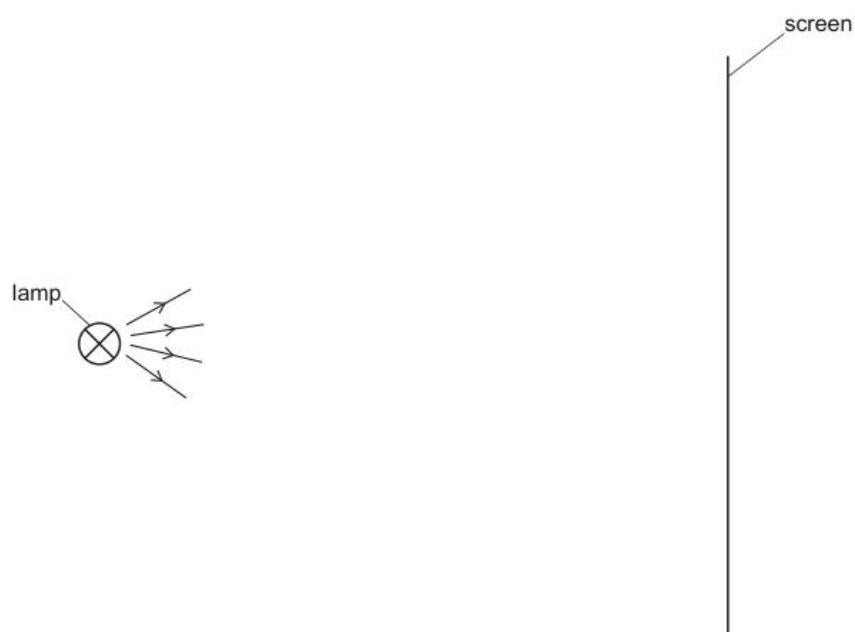


Fig. 10.4

[4]

[Total: 15]

Mark scheme — 5054/22 Q10

5054/22

Cambridge O Level – Mark Scheme
PUBLISHED

May/June 2021

Question	Answer	Marks
9(c)(i)	$(E =) mcT$ numerical or algebraic in any form	C1
	$0.11 \times 830 \times 5$	C1
	460 J	A1
9(c)(ii)	energy output / energy input	C1
	useful energy output / (total) energy input	A1
9(c)(iii)	(energy input =) 460 + 5200 or 5660 (J) seen or 5200 + (energy in 9(c)(i) + 5200)	C1
	92% or 0.92	A1
9(c)(iv)	(thermal) energy <u>lost</u> / heat <u>lost</u> (from battery to environment)	B1
	energy input larger than calculated / 5660 (J) or temperature rise should be larger / is smaller than it should be or thermal energy / heat produced larger than 460 J / calculated	B1

Question	Answer	Marks
10(a)(i)	3 (.0) cm	B1
10(a)(ii)	number of waves per second or 5 / 10 or $f = 1 / T$	C1
	2(.0) Hz	A1
10(a)(iii)	$(v =) f \lambda$ or $s = d / t$ numerical or algebraic in any form	C1
	6(.0) cm / s	A1
10(a)(iv)	frequency stays the same and speed increases	B1

© UCLES 2021

Page 14 of 16

5054/22

Cambridge O Level – Mark Scheme
PUBLISHED

May/June 2021

Question	Answer	Marks
10(a)(v)	> three consecutive incident wavefronts meeting straight refracted wavefronts on the boundary without any wavefronts in-between	B1
	constant wavelength shown between all refracted wavefronts	B1
	all wavefronts refracted in correct direction, i.e. lines between the vertical and the normal	B1
10(b)(i)	four colours from violet, indigo, blue, green, yellow, orange, red	B1
	correct order of four of the above colours	B1
10(b)(ii)	prism	B1
	slit (to produce narrow beam)	B1
	dispersion shown at first or second face, i.e. a ray becoming at least two rays at least one face	B1
	correct refraction on first and second faces for all rays shown	B1

Question	Answer	Marks
11(a)	electron	B1
11(b)	prevents exposure to radiation / for safety (of experimenter and others)	B1
	beta particles stopped / absorbed by metal / case or prevents radiation travelling in all directions / to surroundings	B1
11(c)(i)	decreases (slightly) / little change	B1
	particles spread out or <u>some</u> stopped / absorbed (by air)	B1
11(c)(ii)	no particles detected or <u>large</u> decrease	B1
	(>1m) <u>air</u> stops (beta) particles / <u>air</u> atoms / particles become ionised	B1

© UCLES 2021

Page 15 of 16

2022

May/June 5054/22

Question 8

- 8 (a) Fig. 8.1 shows a ripple tank and the crests of the water wave that is produced in it.

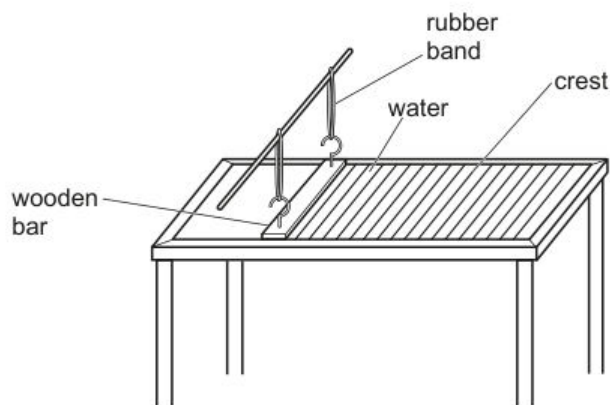


Fig. 8.1

The frequency of the water wave is 2.0 Hz and its amplitude is 3.0 mm .

- (i) Calculate the number of crests produced in 1.5 s .

number = [1]

- (ii) The height of the wave is measured from the level of the undisturbed surface of the water. The height of the wave at one point is 0 at time = 0.

On Fig. 8.2, draw a graph to show how the height of the wave at this point varies with time.

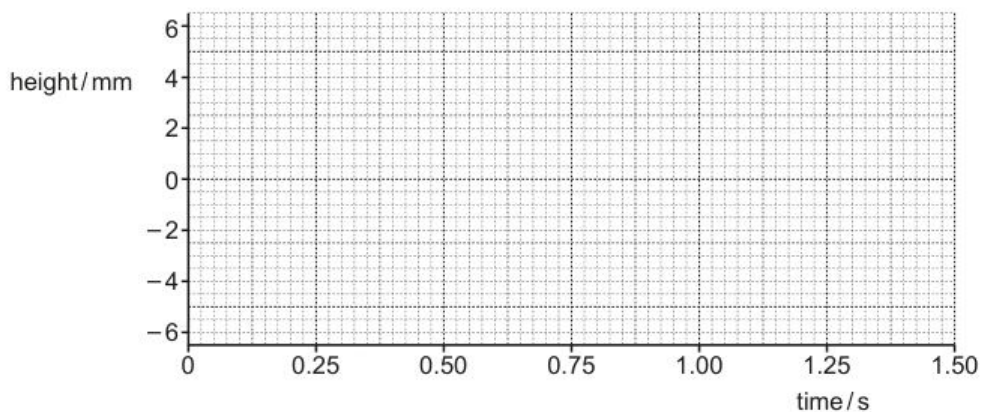


Fig. 8.2

[2]

- (b) (i) The frequency of the wave is increased.

Describe how the apparatus shown in Fig. 8.1 is adjusted so that the frequency of the wave is increased.

.....
 [1]

- (ii) State what happens to the speed and wavelength of the wave as the frequency increases.

speed

wavelength [2]

- (c) The apparatus shown in Fig. 8.1 can be used to demonstrate refraction.

- (i) State the additional apparatus needed to demonstrate refraction.

..... [1]

- (ii) Draw on Fig. 8.3 to show the refraction of the water wave.

Label a boundary where the refraction occurs.



Fig. 8.3

[3]

- (d) Fig. 8.4 shows a connection to the internet made from a remote station A using a satellite above the Earth's surface.

Data is sent between stations A and B using microwaves which travel to and from the satellite.

Station B is connected directly to the internet using optical fibre.

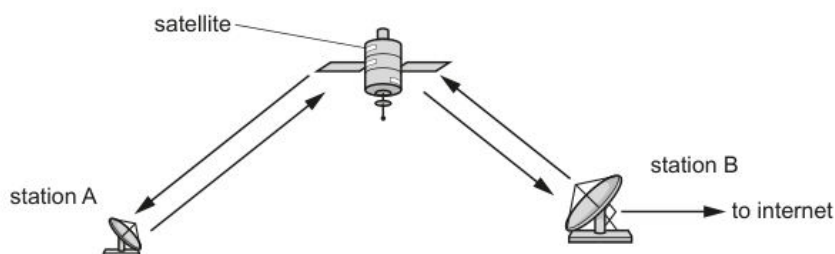


Fig. 8.4 (not to scale)

The speed of microwaves is 3.0×10^8 m/s.

Stations A and B are each 560 km from the satellite.

- (i) Calculate the time taken for data to travel from A to B using microwaves.

time = [3]

- (ii) The same data can be sent from A to B in a shorter time along the surface of the Earth using infrared waves to carry the information in optical fibres.

State **one** property that infrared waves and microwaves have in common.

..... [1]

- (iii) Suggest **one** other advantage of using optical fibres to connect stations A and B directly.

..... [1]

[Total: 15]

Question 9 begins over the page.

Mark scheme — 5054/22 Q8

5054/22

Cambridge O Level – Mark Scheme
PUBLISHED

May/June 2022

Question	Answer	Marks
8(a)(i)	3	B1
8(a)(ii)	shape approx. sinusoidal with amplitude 3 mm for the first complete wave (may then change)	B1
	1 wave taking 0.5 s for first wave shown within half a square or 1.5 s shown for the number of waves in (a)(i)	B1
8(b)(i)	move bar up and down more often (per second) / faster or make rubber band(s) strong(er) / short(er) / tight(er) / more in number or use bar of less mass / weight	B1
8(b)(ii)	<i>speed</i> remains the same	B1
	<i>wavelength</i> decreases	B1
8(c)(i)	(piece of) plastic / (piece of) glass / block / plate or a shallow(er) / deep(er) area / boundary	B1
8(c)(ii)	diagram showing within the ripple tank: crests parallel to bar hitting boundary with a boundary labelled	B1
	correct refraction of plane crests either towards or away from normal on the same side of the normal or no direction change if crests are initially parallel to boundary but with a wavelength change	B1
	different wavelength shown before and after refraction and constant before and after (by eye)	B1
8(d)(i)	($t =$) distance / speed algebraic or numerical seen in any form	C1
	$2 \times 560(000)$ seen or used or 0.0019 (s)	C1
	0.0037 s	A1

© UCLES 2022

Page 10 of 12

Oct/Nov 5054/22

Question 4

4 A sound wave, travelling in air or water, contains compressions and rarefactions.

- (a) Describe an experiment to show that a medium is required to transmit sound waves. You may draw a labelled diagram if you wish.

.....

.....

.....

.....

..... [3]

- (b) When a sound wave passes through air or water, the molecules of the air or water move.

Describe the motion of the molecules due to the sound wave.

.....

.....

..... [2]

- (c) Describe what is meant by 'a rarefaction'.

.....

..... [1]

(d) Fig. 4.1 shows a dolphin.

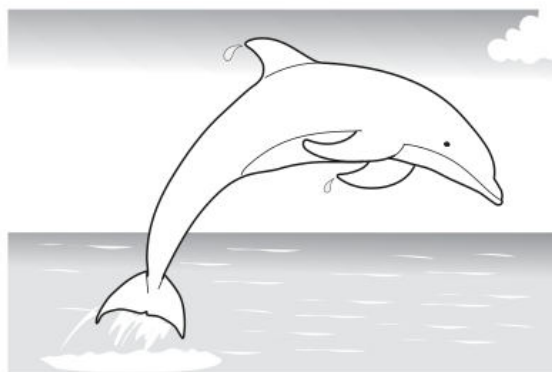


Fig. 4.1

The range of audible sound frequencies for the dolphin is 75 Hz to 140 000 Hz.

In water, the speed of sound is 1500 m/s.

Calculate the smallest wavelength of sound in water that the dolphin can hear.

wavelength = [2]

[Total: 8]

GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>'List rule' guidance</u> For questions that require n responses (e.g. State two reasons ...): - The response should be read as continuous prose, even when numbered answer spaces are provided. - Any response marked <i>ignore</i> in the mark scheme should not count towards n. - Incorrect responses should not be awarded credit but will still count towards n. - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should not be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. - Non-contradictory responses after the first n responses may be ignored even if they include incorrect science.

Question	Answer	Marks
3(a)	chemical potential energy and no other energy	B1
3(b)(i)	radiation or infrared mentioned	B1
	radiation and infrared (travel to the man)	B1
	energy / infrared / radiation <u>absorbed</u> by the man / pullover or travels in straight line or travels at the speed of light or increases (man's) internal energy	B1
3(b)(ii)	(shiny surface) <u>reflects</u> radiation (to man)	B1
	less (thermal) energy escapes or more (thermal) energy reaches / absorbed by man	B1
3(b)(iii)	EITHER black (surface) is good absorber / poor reflector of infrared radiation	B1
	more (thermal) energy absorbed or more thermal energy transferred to man	B1
	OR trapped air and reduced convection / air poor conductor / air is a good insulator	(B1)
	less (thermal) energy lost	(B1)

Question	Answer	Marks
4(a)	equipment e.g. (evacuable) container and (suspended electric) bell / source of sound and (air) <u>pump</u>	B1
	action e.g. ringing / sound (produced) / sound heard and evacuate air / vacuum (produced)	B1
	observation e.g. transmission / hearing of sound ceases / no sound	B1

2023

May/June 5054/21

Question 7

- 7 A water wave in a ripple tank diffracts as it passes through the gap in a barrier.

Fig. 7.1 shows a drawing made by a student of the crests in the pattern.

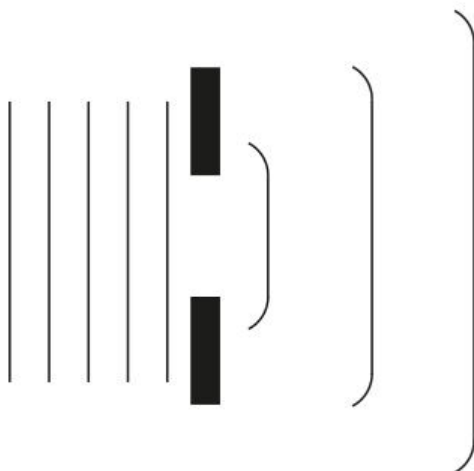


Fig. 7.1

- (a) State **one** way in which the student's drawing of the crests is wrong.

.....
 [1]

- (b) The gap in the barrier is now made smaller than the wavelength, as shown in Fig. 7.2.

Complete Fig. 7.2 with at least three crests to show the new diffraction pattern.

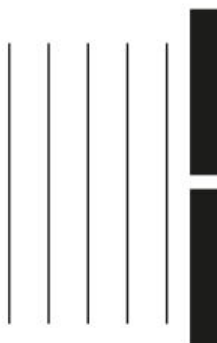


Fig. 7.2

[2]

- (c) In one part of the ripple tank, the water wave has a speed of 6.0 cm/s and a wavelength of 2.0 cm .

The wave then passes into a shallower region of the tank.

The speed of the wave in the shallow region is 4.0 cm/s .

- (i) Define the term 'wavelength'.

.....
 [1]

- (ii) Calculate the frequency of the wave.

frequency = Hz [2]

- (iii) Calculate the wavelength of the wave in the shallow part of the tank.

wavelength = cm [1]

[Total: 7]

Mark scheme — 5054/21 Q7

5054/21

Cambridge O Level – Mark Scheme
PUBLISHED

May/June 2023

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

© UCLES 2023

Page 5 of 13

5054/21

Cambridge O Level – Mark Scheme
PUBLISHED

May/June 2023

Question	Answer	Marks
7(a)	wavelength has altered / too large or crests in shadow should be bent / quarter circular above or below gap or vertical lines too straight on two crests	B1
7(b)	at least 3 semicircles centered on the gap	B1
	wavelength constant and the same on right and left of gap and some bending	B1
7(c)(i)	distance between two (consecutive) identical points such as two (consecutive) crests	B1
7(c)(ii)	$v = f\lambda$ numerical or algebraic e.g. $f = 6 / 2$ (Hz)	C1
	3(.0) Hz	A1
7(c)(iii)	1.3 (cm)	B1

© UCLES 2023

Page 10 of 13

Question 5

5 Water waves are transverse waves.

(a) Underline **two** other examples of transverse waves.

seismic P-waves

seismic S-waves

sound

X-rays

[1]

(b) Fig. 5.1 shows a wooden bar and a glass block in a ripple tank. The depth of water in the tank is less than the height of the glass block.

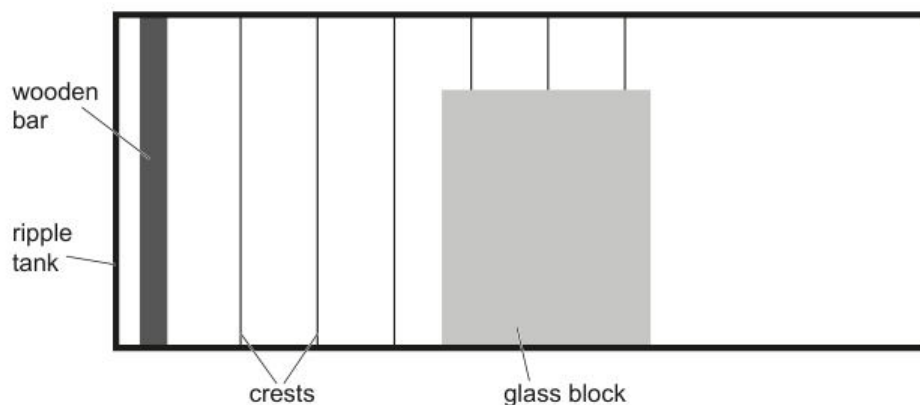


Fig. 5.1 (not to scale)

The wooden bar moves up and down once every 0.15 s to create the crests.

(i) The speed of the water wave is 27 cm/s.

Calculate the frequency and the wavelength of the wave.

frequency = Hz

wavelength = cm
[3]

(ii) The wave diffracts at the right-hand edge of the glass block.

On Fig. 5.1 draw **two** crests after they pass the glass block to show the diffraction. [2]

(iii) Describe how a wave with a smaller wavelength is made with the wooden bar.

.....
..... [1]

(iv) Describe how a decrease in wavelength affects the diffraction.

.....
..... [1]

[Total: 8]

[Turn over]

Mark scheme — 5054/22 Q5

5054/22

Cambridge O Level – Mark Scheme
PUBLISHED

May/June 2023

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	'List rule' guidance For questions that require <i>n</i> responses (e.g. State two reasons): - The response should be read as continuous prose, even when numbered answer spaces are provided. - Any response marked <i>ignore</i> in the mark scheme should not count towards <i>n</i>. - Incorrect responses should not be awarded credit but will still count towards <i>n</i>. - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should not be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. - Non-contradictory responses after the first <i>n</i> responses may be ignored even if they include incorrect science.

© UCLES 2023

Page 4 of 12

5054/22

Cambridge O Level – Mark Scheme
PUBLISHED

May/June 2023

Question	Answer	Marks
4(a)(i)	–273 (°C) and 0 (K)	B1
4(a)(ii)	particles) move fast(er)	B1
	vibrate / oscillate with larger amplitude	B1
4(a)(iii)	internal energy increases	B1
	temperature stays the same	B1
4(b)(i)	($E = mcT$ in any form numerical or algebraic	C1
	8000 (J)	A1
4(b)(ii)	100–44 or 56	C1
	0.48 or 0.47 (J/(g °C))	A1

Question	Answer	Marks
5(a)	seismic S-waves and X-rays	B1
5(b)(i)	(wavelength =) v / f in any form or (distance =) vt or $f = 1 / T$ or $1 / 0.15$	C1
	frequency 6.7 (Hz)	A1
	wavelength 4.0 (cm)	A1
5(b)(ii)	<u>straight</u> wavefronts above block to the right with same wavelength	B1
	correct diffraction – wavefronts quarter circles with centre at top corner	B1
5(b)(iii)	increase frequency of movement of (wooden) bar	B1

© UCLES 2023

Page 8 of 12

2024

May/June 5054/21

Question 4

2

- 1 A ball is released from rest at point A and moves along a smooth track ABCDE as shown in Fig. 1.1. The ball is shown at point A and as it passes point B.

The ball is always in contact with the track and air resistance is negligible.

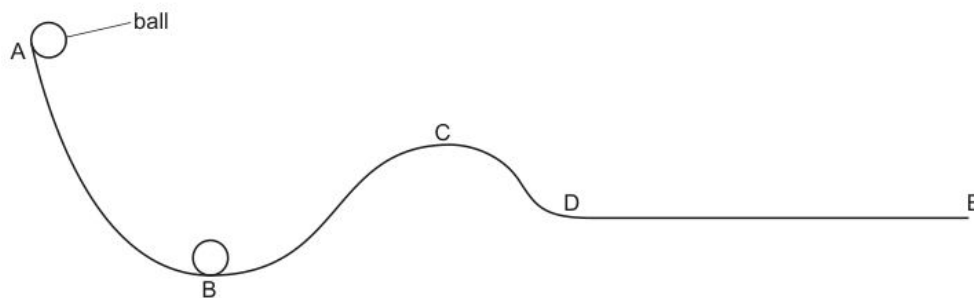


Fig. 1.1

Fig. 1.2 shows the distance–time graph for the ball as it moves from A to E.

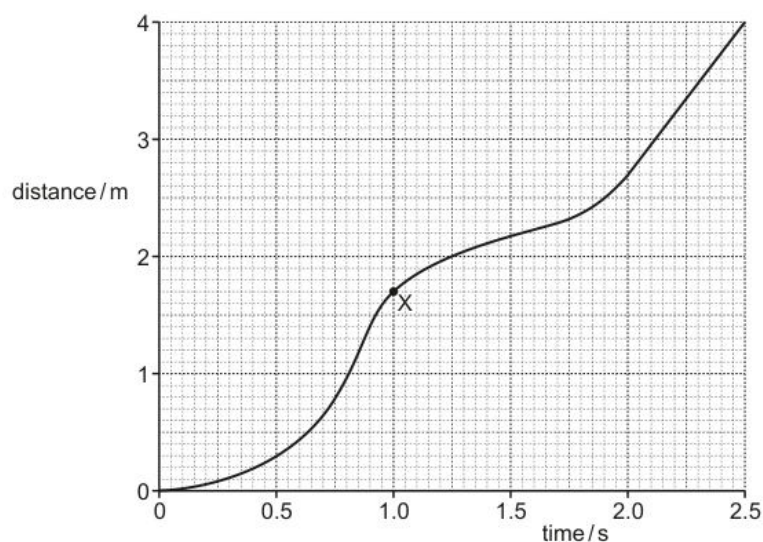


Fig. 1.2

- (a) (i) On Fig. 1.2, mark:

- with the letter P **one** point where the ball accelerates
- with the letter Q **one** point where the ball has constant speed.

[2]

- (ii) Determine the speed of the ball at point X on Fig. 1.2.

Show your working.

speed = m/s [3]

- (b) The speeds of the ball at A, B, C and D are v_A , v_B , v_C and v_D respectively.

Arrange these four speeds from slowest to fastest.

slowest $\longrightarrow$ fastest

--	--	--	--

[1]

- (c) Work is done to transfer energy between energy stores as the ball moves from A to B.

Name the force involved in the work done and describe the energy transfer.

.....

.....

..... [3]

- (d) The track at point B is circular in shape.

On Fig. 1.1, draw an arrow to show the direction of the resultant force on the ball at point B.

[1]

[Total: 10]

- 2 A small aircraft takes off from the horizontal deck of a ship.

Before taking off, the aircraft is held in place by a holdback bar.

When the holdback bar is released, the aircraft is pulled along the deck by a steam-powered piston as shown in Fig. 2.1.

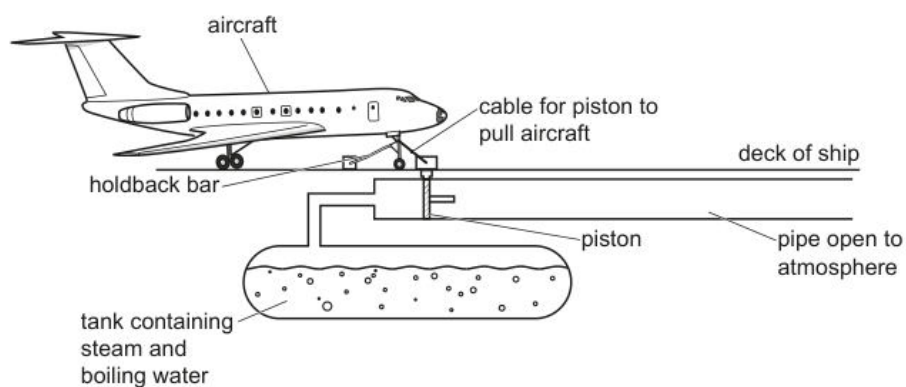


Fig. 2.1 (not to scale)

The steam exerts a high pressure on the piston.

- (a) (i) Explain, using ideas about particles, how the steam creates a pressure on the piston.

.....

.....

.....

..... [2]

- (ii) Explain why the pressure on the piston increases as the temperature of the steam increases.

.....

.....

.....

..... [2]

5

- (b) When the pressure is high enough, the holdback bar is released. The steam pushes the piston along the pipe shown in Fig. 2.1.

The piston has a cross-sectional area of 0.30 m^2 .

The pressure of the steam in the tank is $2.1 \times 10^6 \text{ Pa}$.

Atmospheric pressure is $1.0 \times 10^5 \text{ Pa}$.

Determine the resultant force on the piston caused by the pressure difference.

Show your working.

force = N [2]

- (c) The force calculated in (b) causes the aircraft to accelerate to a maximum speed of 28 m/s from rest.

The mass of the aircraft is $3.0 \times 10^4 \text{ kg}$.

- (i) Calculate the momentum of the aircraft at the maximum speed.

momentum = kg m/s [2]

- (ii) Calculate the time taken for the aircraft to reach the maximum speed.

time = s [2]

[Total: 10]

- 3 A pan containing ice at -15°C is placed on a gas heater as shown in Fig. 3.1.



Fig. 3.1

- (a) Thermal conduction occurs in the metal from which the pan is made.

By referring to particles, describe the process of thermal conduction in a metal.

.....

.....

.....

..... [2]

- (b) Fig. 3.2 shows how the temperature of the contents of the pan varies with time.

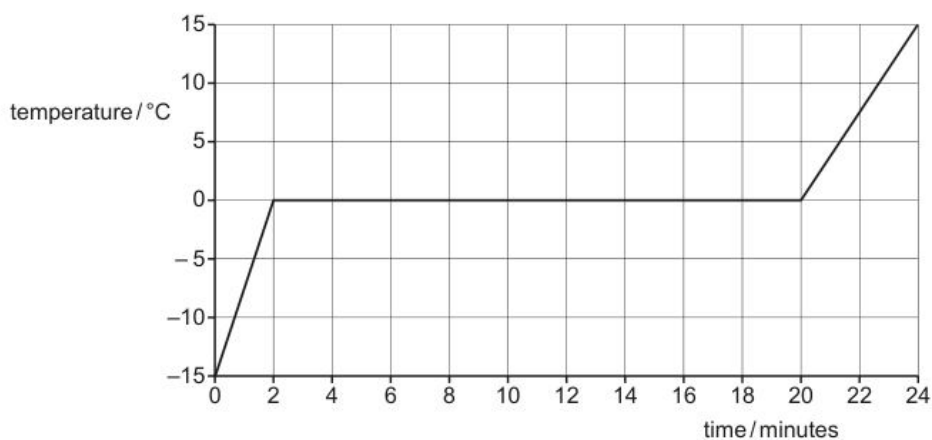


Fig. 3.2

Mark scheme — 5054/21 Q4

5054/21

Cambridge O Level – Mark Scheme
PUBLISHED

May/June 2024

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>'List rule' guidance</u> For questions that require <i>n</i> responses (e.g. State two reasons ...): - The response should be read as continuous prose, even when numbered answer spaces are provided. - Any response marked <i>ignore</i> in the mark scheme should not count towards <i>n</i>. - Incorrect responses should not be awarded credit but will still count towards <i>n</i>. - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should not be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. - Non-contradictory responses after the first <i>n</i> responses may be ignored even if they include incorrect science.

© Cambridge University Press & Assessment 2024

Page 3 of 10

5054/21

Cambridge O Level – Mark Scheme
PUBLISHED

May/June 2024

Question	Answer	Marks
3(b)(v)	(liquid) water has a higher specific heat capacity; water has stronger bonds	B1

Question	Answer	Marks
4(a)(i)	number of (complete) waves (passing a point) per second	B1
4(a)(ii)	$(V =) f \lambda$ in any form	C1
	330 m / s	A1
4(a)(iii)	ANY one from cleaning, prenatal and other medical scanning, sonar, calculation of depth or distance	B1
4(b)(i)	longitudinal wave - oscillation backwards and forwards / in direction of travel of wave	B1
	transverse wave - oscillation at right angles to (direction of travel of) wave	B1
	diagram of each type	B1
4(b)(ii)	earthquake (waves) or other correct source which is a single source that generates both types of wave, e.g. lightning	B1
4(c)	long wavelengths bend around hill and short wavelengths do not bend	B1
	(due to) diffraction	B1

Question	Answer	Marks
5(a)	correct symbol; 	B1
5(b)(i)	correct curvature	B1
5(b)(ii)	(at higher current / voltage) temperature increases	B1
	(at higher current / voltage) resistance increases	B1

© Cambridge University Press & Assessment 2024

Page 7 of 10

Question	Answer	Marks
5(c)	$(I =) V / R$ in any form	C1
	4 (A) or 6 (A) seen or $R_{\text{total}} = 24 (\Omega)$	C1
	10 (A)	A1
5(d)(i)	any integer between and including 12 and 19 (A)	B1
5(d)(ii)	fuse does not melt / blow or circuit does not shut down aaw	B1
	quickly enough or when there is a fault or when the current is large / dangerous	B1
5(d)(iii)	<u>when fuse melts</u> nothing is live / electrocution not possible OR supply of voltage is cut OR voltage is supplied by live wire.	B1

Question	Answer	Marks
6(a)(i)	turns faster / stronger force / more powerful (motor) / larger moment / larger turning effect	B1
6(a)(ii)	correct current direction shown in coil and magnetic field left to right / N to S	B1
6(a)(iii)	(field and current cause) <u>force</u> on left side or <u>force</u> on right side OR coil becomes a magnet	C1
	force upwards on left side and force downwards on right side OR top face of coil is a N pole / bottom face is a S pole	A1
6(b)(i)	circles close to and around each wire	B1
	correct overall shape with stronger field between wires and splaying outwards	B1
	correct direction of field on at least one line and none wrong	B1

Question 5

- 5 Ultrasound is a longitudinal wave which cannot be heard by humans as the frequency of ultrasound is too high.

- (a) Ultrasound waves **cannot** travel in a vacuum.

Explain why.

.....

 [2]

- (b) Describe how a longitudinal wave differs from a transverse wave.

.....

 [2]

- (c) An ultrasound transmitter used for a medical scan produces an ultrasound wave of frequency 8.4 MHz.

In soft human tissue, ultrasound travels at 1500 m/s.

- (i) Calculate the wavelength of this ultrasound wave in soft human tissue.

wavelength = m [3]

- (ii) The ultrasound wave passes from the soft tissue into bone where the speed of the ultrasound wave is greater than 1500 m/s.

State what happens to the frequency and to the wavelength of the ultrasound wave as it passes into the bone.

frequency

wavelength

[2]

[Total: 9]



Mark scheme — 5054/21 Q5

5054/21

Cambridge O Level – Mark Scheme
PUBLISHED

October/November 2024

GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	'List rule' guidance For questions that require <i>n</i> responses (e.g. State two reasons ...): - The response should be read as continuous prose, even when numbered answer spaces are provided. - Any response marked <i>ignore</i> in the mark scheme should not count towards <i>n</i>. - Incorrect responses should not be awarded credit but will still count towards <i>n</i>. - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should not be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. - Non-contradictory responses after the first <i>n</i> responses may be ignored even if they include incorrect science.

© Cambridge University Press & Assessment 2024

Page 3 of 10

5054/21

Cambridge O Level – Mark Scheme
PUBLISHED

October/November 2024

Question	Answer	Marks
4(c)(ii)	frequency increases and wavelength decreases	B1
4(d)	infrared (radiation)	B1

Question	Answer	Marks
5(a)	ultrasound / sound / it is transmitted by <u>vibrating</u> matter / particles	B1
	no matter / particles in a vacuum / no medium	B1
5(b)	(in a longitudinal wave,) vibration (of particles) is parallel to the propagation direction	B1
	in a transverse wave, vibration (of particles) is perpendicular to the propagation direction	B1
5(c)(i)	$(\lambda =) v / f$ or $1500 / 8.4 \times 10^6$	C1
	1.8×10^N	C1
	1.8×10^{-4} (m)	A1
5(c)(ii)	frequency: does not change	B1
	wavelength: increases	B1

© Cambridge University Press & Assessment 2024

Page 7 of 10

2025

May/June 5054/21

Question 5

- 5 (a) Fig. 5.1 is a diagram showing the arrangement of air particles as a longitudinal wave passes through them.



Fig. 5.1

- (i) On Fig. 5.1, mark the centre of a compression with the letter C, and mark the centre of a rarefaction with the letter R. [1]
- (ii) Describe the difference between a compression and a rarefaction.
-
-
- [1]
- (b) In a ripple tank, a water wave is produced by a wooden bar moving up and down on the surface of water.
- (i) The wooden bar makes 45 complete oscillations in 1.0 minute.
- Calculate the frequency of the wave produced.

frequency =Hz [1]

- (ii) The frequency of the water wave is increased by moving the wooden bar up and down more quickly.
- State what happens to the speed and what happens to the wavelength of the wave produced.
- speed
- wavelength [2]





- (iii) The crests of the water wave move into the shallow region shown in Fig. 5.2.

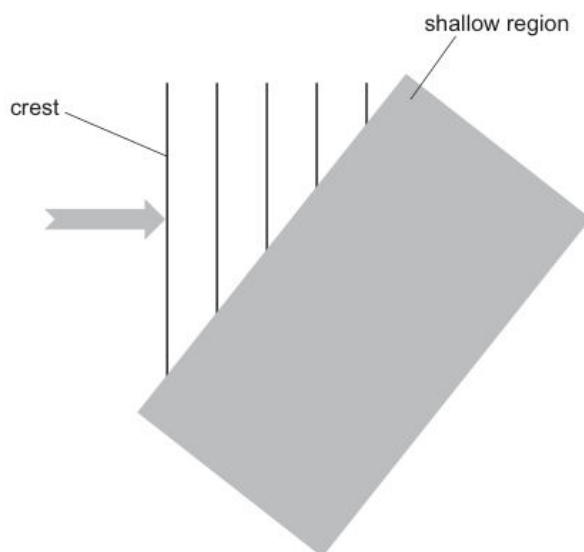


Fig. 5.2

On Fig. 5.2, draw the crests in the shallow region.

[2]

- (c) Describe what is meant by the diffraction of a water wave.

.....

.....

..... [1]

[Total: 8]



Mark scheme — 5054/21 Q5

5054/21

Cambridge O Level – Mark Scheme
PUBLISHED

May/June 2025

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>'List rule' guidance</u> For questions that require n responses (e.g. State two reasons ...): - The response should be read as continuous prose, even when numbered answer spaces are provided. - Any response marked <i>ignore</i> in the mark scheme should not count towards n. - Incorrect responses should not be awarded credit but will still count towards n. - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should not be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. - Non-contradictory responses after the first n responses may be ignored even if they include incorrect science.

© Cambridge University Press & Assessment 2025

Page 3 of 15

5054/21

Cambridge O Level – Mark Scheme
PUBLISHED

May/June 2025

Question	Answer	Marks
5(a)(i)	C and R marked correctly	B1
5(a)(ii)	compressions are where pressure is high / particles close together / density is high	B1
5(b)(i)	0.75 (Hz)	B1
5(b)(ii)	speed: no change	B1
	wavelength: decreases	B1
5(b)(iii)	at least 3 wavefronts in the shallow region parallel to each other and on correct side of normal	M1
	refracted towards the surface on correct side of surface and join wavefronts shown	A1
5(c)	spreading out / bending at an edge / obstacle or spreading out after passing through a (small) gap / hole	B1

© Cambridge University Press & Assessment 2025

Page 11 of 15

Question 7

- 7 Fig. 7.1 shows a simple d.c. motor. The poles at the ends of the magnets are not labelled.

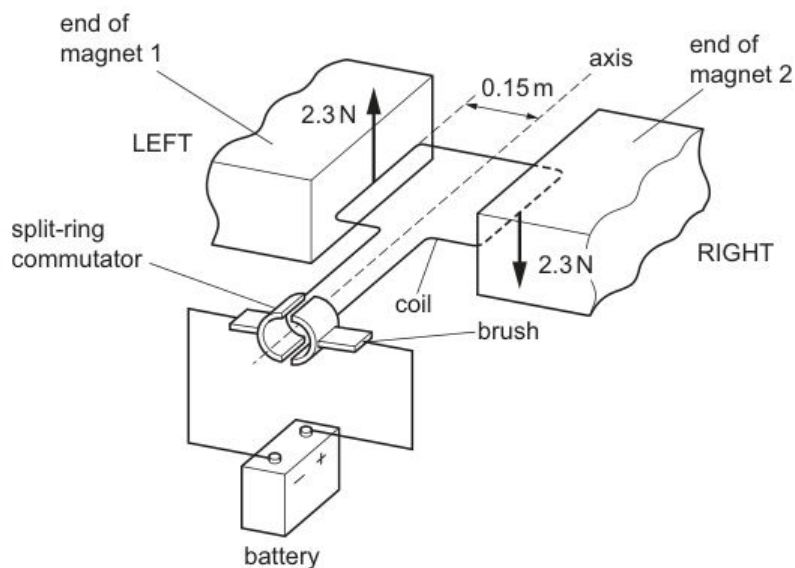


Fig. 7.1

- (a) State the purpose of the split-ring commutator and the brushes.

split-ring commutator

.....

.....

brushes

.....

[3]

- (b) There is an upward force shown on the left side of the coil.

There is a magnetic field in the space between the ends of the magnets.

State the direction of this magnetic field.

Explain how you determined this direction.

direction

explanation

.....

.....

.....

[3]





- (c) State what happens to the rotation of the coil if the magnetic field and the current in the coil are both reversed at the same time.

.....
..... [1]

- (d) In Fig. 7.1, the coil is horizontal. The force upward on the left side and the force downward on the right side are each 2.3 N.

The distance between the sides of the coil and the axis is 0.15 m.

- (i) Calculate the total moment exerted on the coil by these forces.

Give the unit of your answer.

moment =

unit = [3]

- (ii) Explain why the turning effect becomes smaller as the coil becomes more vertical.

.....
.....
..... [1]

[Total: 11]



6	<u>Calculation specific guidance</u> Correct answers to calculations should be given full credit even if there is no working or incorrect working, unless the question states 'show your working'. For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values. For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme. Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.
7	<u>Guidance for chemical equations</u> Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme. State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Question	Answer	Marks
7(a)	<i>split ring commutator:</i> to reverse the current (in the coil)	B1
	reverse current / connections every half turn / twice in one turn / every 180° / when coil is vertical	B1
	<i>brushes:</i> to connect coil / commutator or (split) rings to battery / (external) circuit or to provide current to coil, commutator or (split) rings or to avoid wires (to coil) tangling / twisting	B1
7(b)	<i>direction:</i> (from left) to right / from magnet 1 to magnet 2	B1
	<i>explanation:</i> current is from plus to minus / towards front / out of plane / page / outwards / top to bottom of wire / downwards <u>along wire</u> / towards the split ring on the left	B1
	left-hand rule mentioned and explained using two of current, field or motion / force	B1
7(c)	no change	B1
7(d)(i)	(moment =) force × distance in any form	C1
	0.69	A1
	Nm	B1
7(d)(ii)	(perpendicular) distance between (line of action of) forces decreases or (sides of) coil enter weaker magnetic field or coil leaves magnetic field	B1